

Telerik Web Forms Cryptographic Weakness

Telerik Web Forms Cryptographic Weakness - Progress Telerik, Progress Telerik.

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Content

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Telerik Web Forms Cryptographic Weakness - Telerik UI for ASP.NET AJAX

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[Cryptographic Weakness]()

Updated over 1 year ago

[Problem]()

A third party organization has identified a cryptographic weakness ([CVE-2017-9248](#)) in Telerik.Web.UI.dll that can be exploited to the disclosure of encryption keys (Telerik.Web.UI.DialogParametersEncryptionKey and/or the MachineKey).

[Description]()

Knowledge of these keys in web applications using Telerik UI for ASP.NET AJAX components can lead to:

- cross-site-scripting (XSS) attacks
- leak of MachineKey
-

compromise of the ASP.NET ViewState

-

arbitrary file uploads/downloads

[Solution]()

Update from Jan 5, 2021

Due to the .NET JavaScriptSerializer Deserialization ([CVE-2019-18935](#)) vulnerability, we strongly recommend upgrading to R1 2020 (version 2020.1.114) or later since the patches provided for [CVE-2017-9248](#), [CVE-2017-1135](#), [CVE-2014-2217](#) and [CVE-2017-11317](#) do not prevent it.

Only the upgrade to R1 2020 (2020.1.114) or later can prevent the known vulnerabilities at the time of writing.

You can find more details and instructions at [Allows JavaScriptSerializer Deserialization](#) and [Blue Mockingbird Vulnerability Picks up Steam—Telerik Guidance](#).

Also check the FAQ section at the end of the [Security](#) article.

[Deprecated Solutions]()

Start of the deprecated solutions section:

To ensure your application is not exposed to such a risk, there are three mitigation paths:

- [Use a patch]() for versions between Q1 2013 (2013.1.220) and R2 2017 (2017.2.503)
- [Use a patch]() for some versions between Q1 2011 (2011.1.315) and Q3 2012 SP2 (2012.3.1308)
- If you are on active maintenance, [upgrade to R2 2017 SP1 (2017.2.621) or later]().
- [Prevent access to the Telerik Dialog Handler]()

NOTE: The patches are **not** available on the Telerik NuGet feed.

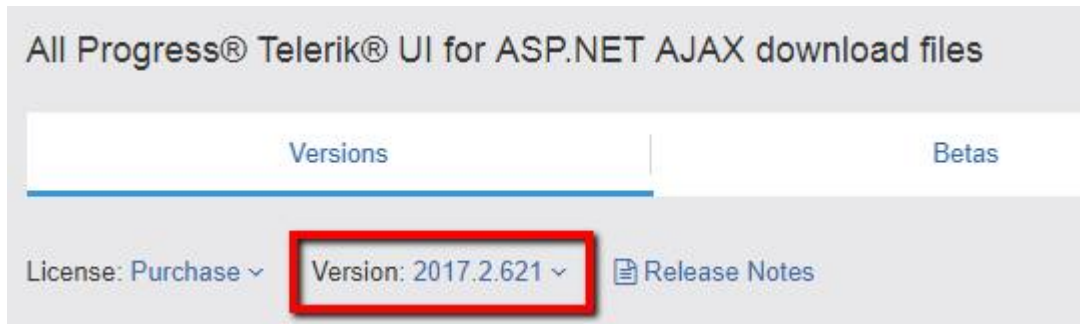
NOTE: If you are targeting .NET 3.5, review the [FIPS Compatibility article](#), because the encryption issue it describes also pertains to these patches.

[Use a patch for versions between Q1 2013 (2013.1.220) and R2 2017 (2017.2.503)]()

Download a patched version from your Telerik.com account **after the 26th of June 2017**:

- Go to your [telerik.com account](#).

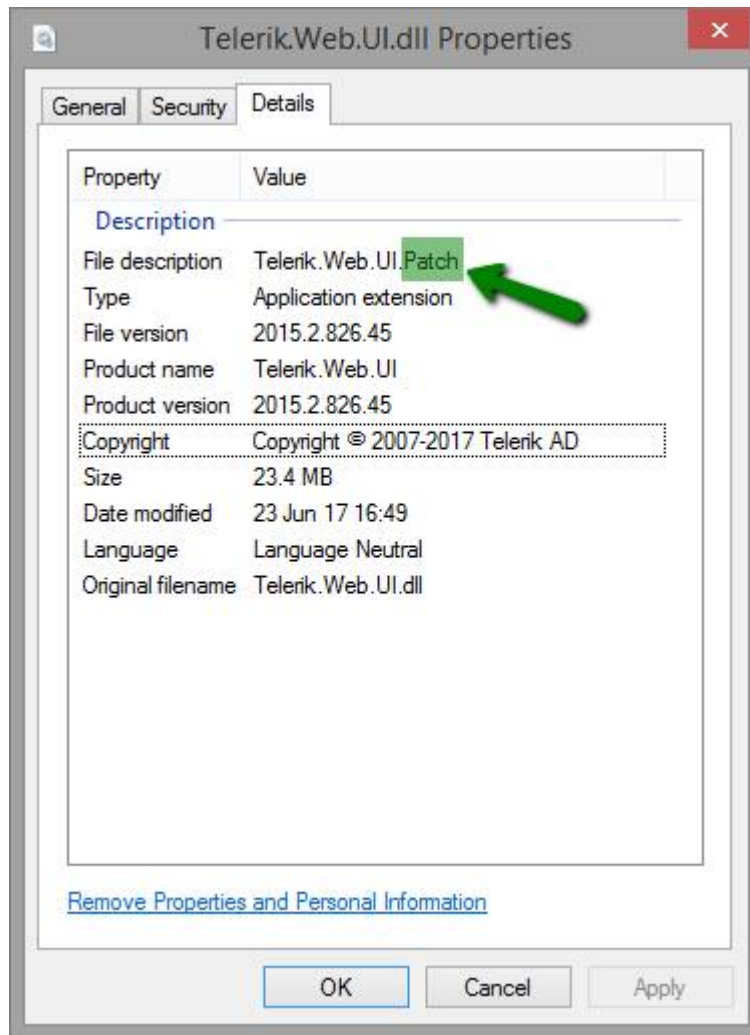
- From the Version dropdown, select your release:



- Download the `SecurityPatch_<your_version>.zip` file.
- [Replace the Telerik.Web.UI assembly in your application](#) with the one of the same >version that you just downloaded.
- Generate **new unique keys** for **Telerik.Web.UI.DialogParametersEncryptionKey** and **MachineKey** in your **web.config**. You can [use the IIS MachineKey Validation Key generator](#) to get them (make sure to avoid the ,IsolateApps portion).

The patched version shows "Telerik.Web.UI.Patch" in the File Description under Properties in Windows Explorer:

How to spot a patched version of Telerik.Web.UI.dll:



Source code for building a **patched** version and [protecting the Telerik.Web.UI assembly](#) is available after **14 Jul 2017**.

[Use a patch for versions between Q1 2011 (2011.1.315) and Q3 2012 SP2 (2012.3.1308)]()

If you are on active maintenance, **upgrade at least to Q1 2013 (2013.1.220)** and follow the same approach for [Using a patch for versions between Q1 2013 (2013.1.220) and R2 2017 (2017.2.503)]().

Due to technical feasibility, the following versions do **not** have patches for this issue:

- Q1 2011 SP2 (2011.1.519)
- Q2 2011 SP1 (2011.2.915)
- Q3 2011 SP1 (2011.3.1305)
- Q1 2012 SP1 (2012.1.411)
- Q2 2012 SP2 (2012.2.912)

If your version lists a `SecurityPatch_<your_version>.zip` file, you can follow the same approach for [Using a patch for versions between Q1 2013 (2013.1.220) and R2 2017 (2017.2.503)]().

[Upgrade to R2 2017 SP1 (2017.2.621) or later.]()

- ****Upgrade**** your Telerik UI for ASP.NET >AJAX version to **R2 2017 SP1** (2017.2.621) or later.
 - **Generate new keys** for **Telerik.Web.UI.DialogParametersEncryptionKey** and **MachineKey** in your ****web.config****. You can [use the IIS MachineKey Validation Key generator](#) to get them (make sure to avoid >the ,IsolateApps portion).
-

[Prevent access to the Telerik Dialog Handler]()

An **alternative** to a fix or a patch is to prevent access to the Telerik DialogHandler. Note that this will make it impossible to use Telerik built-in dialogs for RadEditor and RadSpell.

There are different ways to do that, for example:

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Add a firewall rule that rejects traffic to the handler.

-

Add a URL redirect rule that returns an error page instead of the handler. Note that this will merely redirect the requests to a page of your choosing, usually with a 301 status code.

Here is a basic example:

xml

```
<rewrite>
  <rules>
    <rule name="DisableDialogHandler" enabled="true" stopProcessing="true">
      <match url="^Telerik.Web.UI.DialogHandler.*?$" />
      <action type="Redirect" url="not-allowed.aspx" redirectType="Permanent" />
    </rule>
  </rules>
</rewrite>
```

-

Remove the handler from the web.config:

xml

```

<system.web>
  <httpHandlers>

    <add path="Telerik.Web.UI.DialogHandler.aspx" type="Telerik.Web.UI.DialogHandler"
verb="*" validate="false" />

    <add path="Telerik.Web.UI.DialogHandler.ashx" type="Telerik.Web.UI.DialogHandler"
verb="*" validate="false" />
  </httpHandlers>
</system.web>

<system.webServer>
  <handlers>

    <remove name="Telerik_Web_UI_DialogHandler.aspx" />

    <add name="Telerik_Web_UI_DialogHandler.aspx"
path="Telerik.Web.UI.DialogHandler.aspx" type="Telerik.Web.UI.DialogHandler" verb="*"
preCondition="integratedMode" />

    <add name="Telerik_Web_UI_DialogHandler.aspx"
path="Telerik.Web.UI.DialogHandler.ashx" type="Telerik.Web.UI.DialogHandler" verb="*"
preCondition="integratedMode" />
  </handlers>
</system.webServer>

```

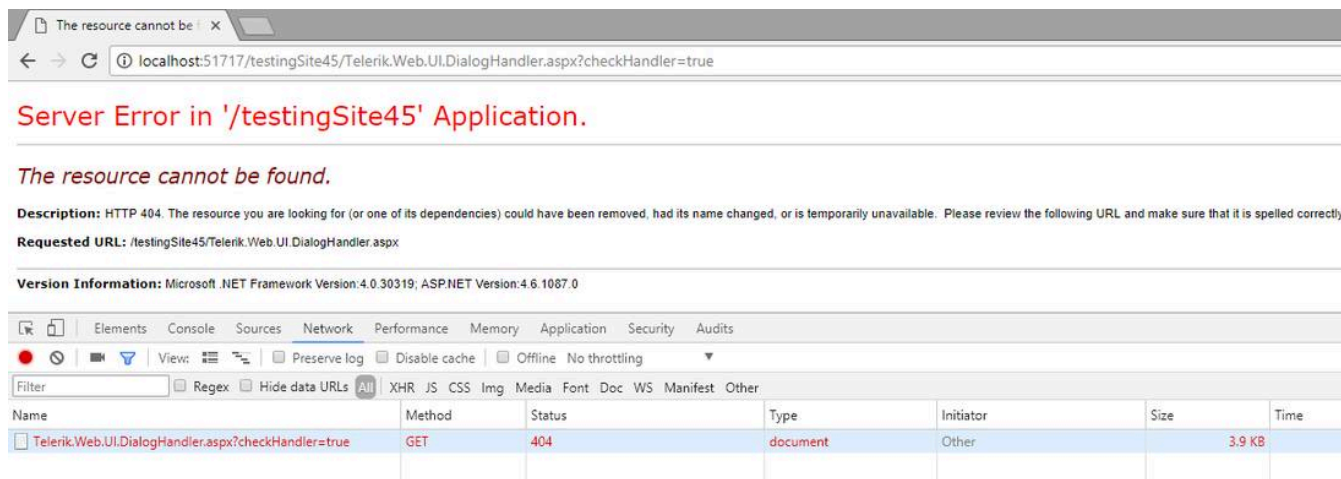
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For **SharePoint** sites, **delete** the **Telerik.Web.UI.SpellCheckHandler.ashx** and **Telerik.Web.UI.DialogHandler.aspx** files that correspond to these handlers. You can find them in the following folders:

- SharePoint 2010: C:\Program Files\Common Files\Microsoft Shared\Web Server Extensions\wpresources\RadEditorSharePoint\6.x.x.0__1f131a624888eed\Resources
- SharePoint 2013: C:\Program Files\Common Files\Microsoft Shared\Web Server Extensions\wpresources\RadEditorSharePoint\7.x.x.0__1f131a624888eed\Resources
- SharePoint 2016: C:\Program Files\Common Files\Microsoft Shared\Web Server Extensions\wpresources\RadEditorSharePoint\7.x.x.0__1f131a624888eed\Resources

You can test whether the handler is available by requesting the following URL under your application root: **Telerik.Web.UI.DialogHandler.aspx?checkHandler=true**

When the handler is not available, you will get an error similar to the following:



End of the deprecated solutions section

[Notes]()

We would like to thank Erlend Leiknes, security consultant in Mnemonic AS, and Thanh Van Tien Nguyen, for responsibly disclosing this vulnerability to us and helping in its resolution.

[External References]()

[CVE-2017-9248](#)

[See Also]()

- [Security](#)
- [RadEditor Security article](#)
- [Allows JavaScriptSerializer Deserialization](#)
- [RadAsyncUpload Security article](#)
- [Unrestricted File Upload/details/unrestricted-file-upload\)](#)
- [Insecure Direct Object Reference/details/insecure-direct-object-reference\)](#)
- [Blue Mockingbird Vulnerability Picks up Steam—Telerik Guidance blog post \(CVE-2019-18935\)](#)

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